

Bilal Amin

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EDUCATION

Universität des Saarlandes

Master's degree, Data Science and AI
Grade: 1.9 (German grading system)

Apr. 2024 – Present

Saarbrücken, Germany

University of Bergen (Erasmus+ Exchange Semester)

Exchange Student, Computer Science

Jan. 2026 – Jun. 2026

Bergen, Norway

National University of Sciences & Technology

Bachelor's in Mechatronics Engineering (Computer Science)
Grade: 1.7 (German grading system)

Sep. 2017 – Sep. 2021

Islamabad, Pakistan

EXPERIENCE

Work Student – Internet Architecture Group

Max Planck Institute for Informatics (MPI)

Oct. 2024 – Mar. 2025

Saarbrücken, Germany

- Designed an automated framework to evaluate congestion control algorithms in NS-3 (Network Simulator).
- Created a user-friendly dashboard to visualize performance metrics under different network conditions.
- Developed system tests and metrics for performance evaluation across varied scenarios.
- Analyzed simulation results to identify optimization opportunities and guide design decisions.

Data Analyst

Convergent Business Technologies

October. 2023 – March. 2024

Islamabad, Pakistan

- Developed reproducible data science workflows using Python and SQL to manipulate and analyze data.
- AS part of training took and finished the CS50 course and made several dashboards regarding it.

Web Development Intern

CureMD

June 2023 – August 2023

Lahore, Pakistan

- Designed user interfaces using HTML, CSS, and JavaScript, improving usability and accessibility for end users.
- Developed APIs in the .NET framework, enhancing data integration and tool development.

Technical Intern

STEP (Space Technology Education and Popularization)

August 2021 - October 2021

Islamabad, Pakistan

- Led workshops teaching programming fundamentals using Scratch, focusing on hands-on learning for underprivileged youth.
- Created educational materials to simplify complex technology concepts, fostering interest in STEM fields.

PROJECTS

AI Learning Assistant using LLMs (RAG)

- Developing an AI-powered learning assistant using Python, LangChain, Streamlit, and FastAPI.
- Integrated a locally hosted Llama model with Retrieval-Augmented Generation (RAG) to answer questions from uploaded PDF documents.
- Implemented document processing, vector embeddings, and ChromaDB for efficient knowledge retrieval and citation-based responses.
- Designed prompt engineering workflows to generate summaries, quizzes, and study materials from course documents.

Stroke Prediction using Machine Learning

- Developed a predictive model to assess stroke risk based on patient data, achieving 85% accuracy.
- Leveraged Python libraries such as Scikit-Learn and Pandas to process and analyze medical datasets.

MarketPulse Website Design and HCI Analysis

- Designed and developed a responsive website for MarketPulse.
- Applied key HCI principles such as Mapping, Consistency, Simplicity, Feedback, Visibility, Constraints, and Mental Models to enhance usability and user experience.
- Focused on clear information architecture, accessible design, and engaging user interaction patterns to support better product usability.

MOOCs

- Machine Learning Specialization by Andrew Ng
- Google Data Analytics Professional Certificate (8 of 12)
- Python for Data Science, AI & Development by IBM
- CS50's Introduction to Artificial Intelligence with Python by Harvard University

TECHNICAL SKILLS AND INTERESTS

Programming: Python, JavaScript, SQL, HTML, CSS, C++, C#, Java

Machine Learning & AI: Scikit-Learn, Large Language Models (LLMs), Prompt Engineering, Retrieval-Augmented Generation (RAG)

LLM Frameworks: LangChain, Ollama, ChromaDB

Web Frameworks: FastAPI, Streamlit, ReactJS

Libraries: Pandas, NumPy, Matplotlib

Tools & Platforms: GitHub, GitHub Actions, VS Code, Jupyter, Tableau, Figma, Canva, Wix

Interests: Artificial Intelligence, LLM Applications, Generative AI, Intelligent Learning Systems, Workflow Automation, Data Visualization